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Distinguishing between metabolically active and dormant bacteria on paper

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Abstract Switching between metabolically active and dormant states provides bacteria with protection from environmental stresses and allows rapid growth under favorable conditions. This rapid growth can be detrimental to the environment, e.g., pathogens in recreational lakes, or to industrial processes, e.g., fermentation, making it useful to quickly determine when the ratio of dormant to metabolically active bacteria changes. While a rapid increase in metabolically active bacteria can cause complications, a high number of dormant bacteria can also be problematic, since they can be more virulent and antibiotic-resistant. To determine the metabolic state of *Escherichia coli* and *Salmonella* Typhimurium, we developed two paper-based colorimetric assays. The color changes were based on oxidoreductases reducing tetrazolium salts to formazans, and alkaline phosphatases cleaving phosphates from nitrophenyl phosphate salt. Specifically, we added iodophenyl-nitrophenyl-phenyl tetrazolium salt (INT) and methylphenazinium methyl sulfate to metabolically active bacteria on paper and INT and para-nitrophenyl phosphate salt to dormant bacteria on paper. The color changed in less than 60 min and was generally visible at 10^3 CFU and quantifiable at 10^6 CFU. The color changes occurred in both bacteria, since oxidoreductases and alkaline phosphatases are common bacterial enzymes. On one hand, this feature makes the assays suitable to a wide range of applications, on the other, it requires specific capture, if only one type of bacterium is of interest. We captured *Salmonella* or *E. coli* with immobilized P22 or T4 bacteriophages on the paper, before detecting them at levels of 10^2 or 10^4 CFU, respectively. Determining the ratio of the metabolic state of bacteria or a specific bacterium at low cost and in a short time, makes this methodology useful in environmental, industrial and health care settings.

Keywords Bacteriophage · *Escherichia coli* · *Salmonella* · Metabolic state · Detection · Paper

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Introduction

A large fraction of bacteria exist in a dormant state, including over 30% in sludge, 83% in soil, and 50% or more in lakes (Lennon and Jones 2011). When dormant bacteria change to an active metabolic state, their sometimes rapid reproduction can lead to a bacterial bloom. In a water body, a bloom can cause anoxic conditions in the environment, resulting in a fish kill (Hallegraeff 1993), while in an industrial setting, it could potentially alter the yield and product properties during fermentation (Murphree et al. 2014). Rapidly reproducing pathogenic bacteria can cause disease outbreaks (Coates 2003), especially since dormancy can increase pathogenicity (Coates 2003; Poncet et al. 2009) and antibiotic resistance (Becker et al. 2006; Lewis 2007). To prevent, or at least

counteract these effects, we need methods to determine changes in the ratio of dormant and metabolically active bacteria quickly and preferably techniques that can be applied directly in the field.

When the metabolic state of bacteria changes, the type and level of enzymes expressed change (Ayyash et al. 2014; Gómez et al. 2002). For example, dormant *Escherichia coli* express more alkaline phosphatases (ALP) and oxidoreductases associated with survival, such as nitrate reductase (Yang and Metcalf 2004; Clegg et al. 2006), compared to metabolically active *E. coli*, which express less ALP (Achbergerová et al. 2011), and more oxidoreductases in their central metabolism (Almaas et al. 2004). A range of viable but non-culturable (VBNC) bacteria (Li et al. 2014) show similar behavior including *Campylobacter jejuni*, *Helicobacter pylori* (Jones 2001), and *Salmonella* (Becker et al. 2006). Both oxidoreductases and alkaline phosphatases react with dyes (Blake et al. 1984), including tetrazolium salts that reduce to formazans for oxidoreductases (Rodriguez et al. 1992; Berridge et al. 2005) and nitrophenyl phosphates for ALP (Bessey et al. 1946). Tetrazolium salts are commonly used to determine the number of viable cells in solutions (Berridge et al. 2005), since both the kinetics of the color change and the final color intensity are related to the concentration of the enzymes (Blake et al. 1984).

Colorimetric detection of bacteria in solution by simply adding relevant dyes is very common (Blake et al. 1984), but often requires instrumentation for the color analysis. On materials, on the other hand, the color is localized and with that more concentrated allowing for analysis by eye and making the assays more portable (Pelton 2009; Anany et al. 2011a). Paper, an easy to modify and low-cost material (Martinez et al. 2007; Yetisen et al. 2013; Hu et al. 2014), has been used to colorimetrically detect liver function enzymes and small molecules in blood (Nie et al. 2010; Vella et al. 2012; Hu et al. 2014), and bacteria contamination in food (Jokerst et al. 2012), beverages (Rompré et al. 2002), and water (Fiksdal et al. 1994; Lazcka et al. 2007). These assays focus on identifying specific bacteria in a matrix, by using reactive dyes interacting with enzymes present in only those bacteria (Adkins et al. 2017).

In this study, we used tetrazolium salts and nitrophenyl phosphate to determine the metabolic state of *E. coli* and *Salmonella* Typhimurium on paper. Since the reactive dyes were not specific for the bacteria, we combined the assays with bacteriophages adsorbed on the paper to capture only bacteria of interest. Bacteriophages adsorbed (Cademartiri et al. 2010) on materials or chemically bound to them (Wang et al. 2016) have been used to capture *E. coli* (Anany et al. 2011) and *Salmonella* (Lakshmanan et al. 2007; Li et al. 2010) before detection. We used T4 bacteriophages for *E. coli* and P22 bacteriophages for *Salmonella*. Bacteria adsorbed on the paper or captured by immobilized bacteriophages were detected with iodophenyl-nitrophenyl-phenyl tetrazolium salt (INT) mixed with an electron transporter at neutral pH or nitrophenyl phosphate at alkaline pH. The assay at neutral pH was similar to those used for metabolically active bacteria in solution (Berridge et al. 2005) and showed a stronger purple color for metabolically active bacteria on paper. The assay at alkaline pH, on the other hand, exhibited a stronger purple color for dormant bacteria. The colors were analyzed visually and quantified by image analysis allowing us to relate the color intensity to the metabolic state and concentration of *E. coli* and *Salmonella* on the paper.

Materials and methods

Bacteria and phage cultures

Salmonella enterica subspecies *enterica* ser. Typhimurium (ATCC 19585) and *E. coli* K12 (ATCC 25404) were purchased from the American Type Culture Collection (Manassas, VA), while environmental *E. coli* (*E. coli* LLV) were isolated from lake water (Lake LaVerne, Ames, IA) by plating the water on modified mTEC agar. Bacteria were propagated and enumerated (plate counts) using standard techniques with Luria-Bertani (LB) broth and agar. Dormant bacteria were prepared by inoculating filtered autoclaved lake water (FALW) with 10^9 CFU/ml and storing the samples for 30 to 40 days at 4 °C (Ozkanca and Flint 1997). Dormant bacteria were enumerated using plate counts, before and after preparation in FALW. About 10% of these bacteria survived storage, which were classified as dormant. T4 (ATCC 11303-B4) and P22 (ATCC 19585-B1) bacteriophages were propagated and enumerated on overlays (Clokie and Kropinski 2009) and stored in lambda buffer (100 mM NaCl, 16.6 mM $MgSO_4$, 50 mM Tris-HCl buffer pH 7.5, and 0.01% *w*/*v* gelatin).

Colorimetric assays for bacterial enzymes on paper

Colorimetric assays on Whatman® grade 3 filter papers were based on the reduction of tetrazolium salts or the hydrolysis of para-nitro-phenyl-phosphate. To create reaction zones, the paper was cut into circles with 38.5 mm^2 surface area, or wax rings with 5-mm diameter were printed with Xerox Color Qube 8570DN™ (Norwalk, CT) and melted through the paper (5–8 min at 85 °C) (Carrilho et al. 2009). The assays were performed by first adding 10 μl of the bacteria in phosphate-buffered saline (PBS) at pH 7 and 5 μl of assay solution to the paper. The assay solution for metabolically active bacteria was a freshly prepared solution with 0 to 2.5 mM 2-(4-iodophenyl)-3-(4-nitrophenyl)-5-phenyl-2H-tetrazolium chloride (INT), and 0 to 1 mM 5-methylphenazin-5-ium methyl sulfate (PMS) in PBS at pH 7 (see Table S1 for detailed ratios). For dormant bacteria, we used a freshly prepared solution of 0

to 3 mM INT and 1 to 45 mM nitrophenyl phosphate (PNPP) in 0.5 mM Tris buffer at pH 10 (details in Table S2). Using 2.5 mM INT and 1 mM PMS, the above was repeated using combined mixtures of metabolically active and dormant *Salmonella* at 10^8 CFU/ml (100% active to 100% dormant bacteria). For all samples, the bacteria and assay reacted in minutes and were dried in the dark (1–4 h) before scanning (Canon LiDE 110), while measurement of the color intensity occurred using ImageJ™ (Vella et al. 2012). The color for INT/PMS mixtures was measured in the green channel while INT/PNPP mixtures were analyzed in the blue channel, since these led to the strongest differences at difference concentrations of bacteria. At least five replicates of each test were performed.

Immobilization of bacteriophages and capture of bound bacteria

Bacteriophages were immobilized on paper by adsorption. Single circles of Whatman® grade 3 filter papers were placed vertically inside a 96-well plate, and 200 μl of bacteriophage (10^{10} PFU/ml) were added, completely covering the paper on both sides. After 48 h at room temperature in the dark, the bacteriophage solution was removed and unreacted sites on the paper were blocked by immersing the papers in 200 μl of 1 *w*/*v*% bovine serum albumin (BSA) for 1 h. The blocked papers were washed with 100 μl of PBS buffer before use. *E. coli* (10^3–10^7 CFU) or *Salmonella* (10^2–10^6 CFU) were added and allowed to interact for 5 min at 37 °C and shaking at 300 RPM. Unbound bacteria were removed by four washes with 100 μl of PBS buffer. The captured bacteria were analyzed as described above. Eight replicates were performed for each test.

Results

Development of colorimetric assays

Each metabolic state required its own assay on paper. Oxidoreductases in metabolically active bacteria reacted with a tetrazolium salt at neutral pH, while in dormant bacteria, oxidoreductases and alkaline phosphatases reacted with a mixture of a tetrazolium salt and nitrophenyl phosphate at alkaline pH.

Iodonitrotetrazolium chloride for metabolically active *E. coli* on paper

In solution, several tetrazolium salts change color in the presence of bacteria (Berridge et al. 2005). On paper, only INT showed a purple color in the presence of *E. coli*, while methylthiazolyl-diphenyl-tetrazolium bromide and neotetrazolium chloride did not change color. INT developed color faster when combined methylphenazinium methyl sulfate (PMS), an electron transporter. Changing the concentrations and ratios of INT and PMS at constant *E. coli* (5×10^3 CFU/mm^2) did not lead to obvious trends in the developed color (Fig. 1, Table S1). The darkest purple developed for the highest concentration of INT (2.5 mM) and PMS (1 mM), while the second darkest purple was present at the lowest concentrations of INT (0.5 mM) and PMS (0.25 mM). Two other concentrations and ratios of INT and PMS (1.5 mM INT/0.5 mM PMS, and 2 mM INT/1 mM PMS) developed a weaker, but visible color. When images were analyzed digitally, the green component of the purple color exhibited the largest changes with concentrations of *E. coli*, and was used for quantitative analysis. At the highest concentration of INT and PMS, the green component of the color increased linearly with increasing concentrations of *E. coli* on the paper (Fig. 1), with a color change significant from background at 5×10^3 CFU/mm^2.

Para-nitrophenyl phosphate and INT for dormant *E. coli* on paper

The detection of dormant bacteria was based on two classes of enzymes: oxidoreductases and alkaline phosphatases (ALP). These enzymes have been observed to be active in dormant

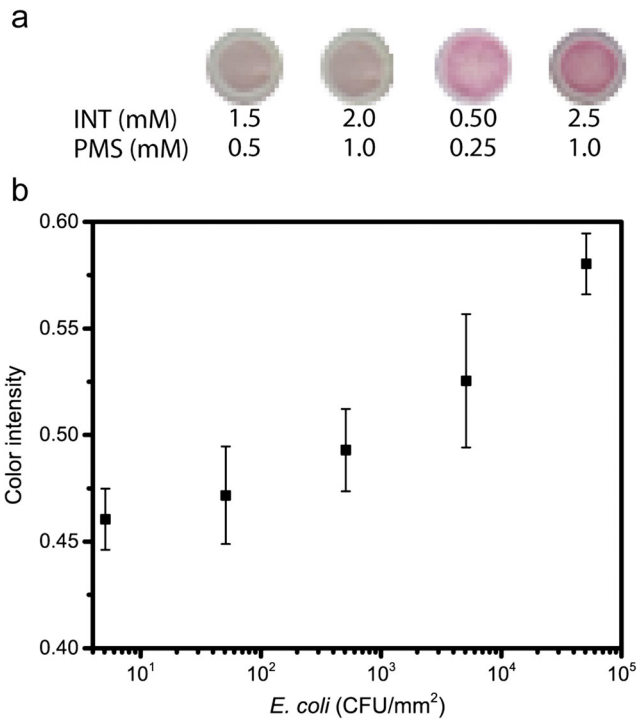

Fig. 1 INT/PMS assay on paper detected metabolically active *E. coli*. **a** Distinct ratios of INT and PMS produced different color intensity values on paper at the same concentration of *E. coli*. **b** Different concentrations of *E. coli* on paper detected by the assay with the highest color intensity 2.5 mM INT and 1 mM PMS. The green component of the purple color was used for analysis. Error bars correspond to two standard errors ($n = 6$)

bacteria. We used INT and para-nitrophenyl phosphate (PNPP) as substrates for these enzymes. PNPP is cleaved by ALP to produce yellow para-nitrophenol while INT is transformed to purple formazan by action of oxidoreductases. Considering that para-nitrophenol was difficult to observe on the white paper, and taking advantage of the presence of both type of enzymes in dormant bacteria, we combined PNPP and INT substrates to detect simultaneously ALP and oxidoreductases activities. The combined substrate INT/PNPP led to a stronger purple color that was easier to distinguish on paper compared to these substrates alone.

For dormant *E. coli* at pH 10, the color of the assay increased with increasing concentration of INT (Fig. 2, Table S2), over a wider range of INT concentrations than at pH 7 due to an increased solubility of INT (3 versus 2.5 mM). Adding PNPP (10 to 45 mM) to a constant concentration of INT (3 mM) led to an increase in color with increasing PNPP concentration as long as 5 mM of PNPP was present (Fig. 2a). The strongest color in the blue channel developed for the highest concentrations of INT/PNPP (3 mM/45 mM), which was used to measure color intensity at different concentrations of dormant *E. coli* (Fig. 2b). There was a visible difference in color between 10^3 and 10^4 CFU/mm^2 (Fig. S1), which led to a significant increase in the color intensity in the blue channel.

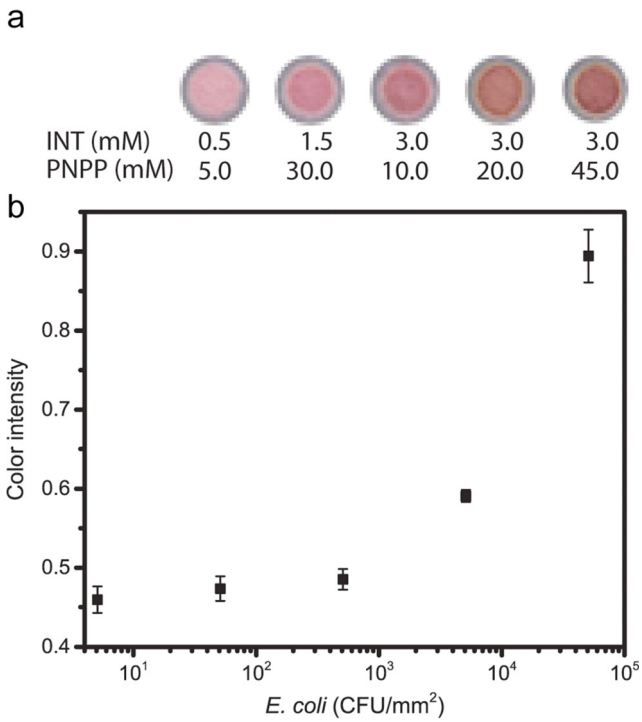

Fig. 2 INT/PNPP assay on paper detected dormant *E. coli*. **a** Distinct ratios of INT and PNPP produced different color intensity values on paper at the same concentration of *E. coli*. **b** Different concentrations of *E. coli* on paper detected by the assay with highest color intensity 3 mM INT and 45 mM PNPP. The blue component of the purple color was used for analysis. Error bars correspond to two standard errors (n = 6)

Identifying the metabolic states of dormant *E. coli* and *Salmonella* Typhimurium on paper

After developing assays for metabolically active and dormant *E. coli*, we tested both assays on metabolically active and dormant *E. coli* and *Salmonella* at different concentrations. At each concentration and metabolic state, the bacteria were analyzed on separate papers with mixtures of INT (2.5 mM) and PMS (1 mM) at a neutral pH (INT/PMS), or INT (3 mM) and PNPP (45 mM) at alkaline pH (INT/PNPP). Papers were dried completely before analysis (1-4 h), although a color change for the higher bacterial concentrations (5×10^5 CFU/mm^2) occurred within 5 min.

As expected, INT/PMS showed stronger color for metabolically active bacteria, while INT/PNPP exhibited stronger color for dormant bacteria (Fig. 3). The extent of the color difference depended on the bacterium and its concentration. The INT/PMS assay for metabolically active *Salmonella* showed a color different to the background at 5×10^5 CFU/mm^2, while *E. coli* demonstrated a difference at 5×10^2 CFU/mm^2. On the other hand, for dormant bacteria and the INT/PNPP assay, *Salmonella* changed color at 50 CFU/mm^2, compared to 5×10^4 CFU/mm^2 *E. coli*.

The difference in color between the assays for the same metabolic state was generally more distinct at higher concentrations of bacteria (Fig. 3), except for dormant *Salmonella*. For example, for 5×10^2 CFU/mm^2 of metabolically active *E. coli*, the color for INT/PMS was 8% higher than for INT/PNPP, while at 5×10^5 CFU/mm^2, it was 30% higher. The largest difference in color between INT/PMS and INT/PNPP was seen for dormant *E. coli* at 5×10^5 CFU/mm^2, when INT/PNPP showed a 50% stronger color.

After establishing a relation between the color intensity of each assay, and the metabolic state and concentration of bacteria, we measured metabolically active *Salmonella* in a mixture with dormant *Salmonella* (100% active to 100% dormant) at constant bacteria concentration (10^8 CFU/mL). The mixtures on the papers were analyzed with INT/PMS (2.5 mM/1 mM). As expected, INT/PMS exhibited a darker color for mixtures with higher concentrations of metabolically active *Salmonella*. The color decreased linearly ($R^2 = 0.9997$) with decreasing number of metabolically active until 25% of active *Salmonella*, which exhibited the same color as 100% dormant *Salmonella* (Fig. 4). This linear decrease led to 8, 14, and 19% weaker color in the green channel from 100% active *Salmonella* to 25% active *Salmonella*. This difference in color was visible by eye (Fig. 4).

Detection of dormant *E. coli* and metabolically active *Salmonella* after capture by bacteriophages on paper

Detection of specific bacteria with the non-specific INT/PMS and INT/PNPP assays required specific capture of the bacteria

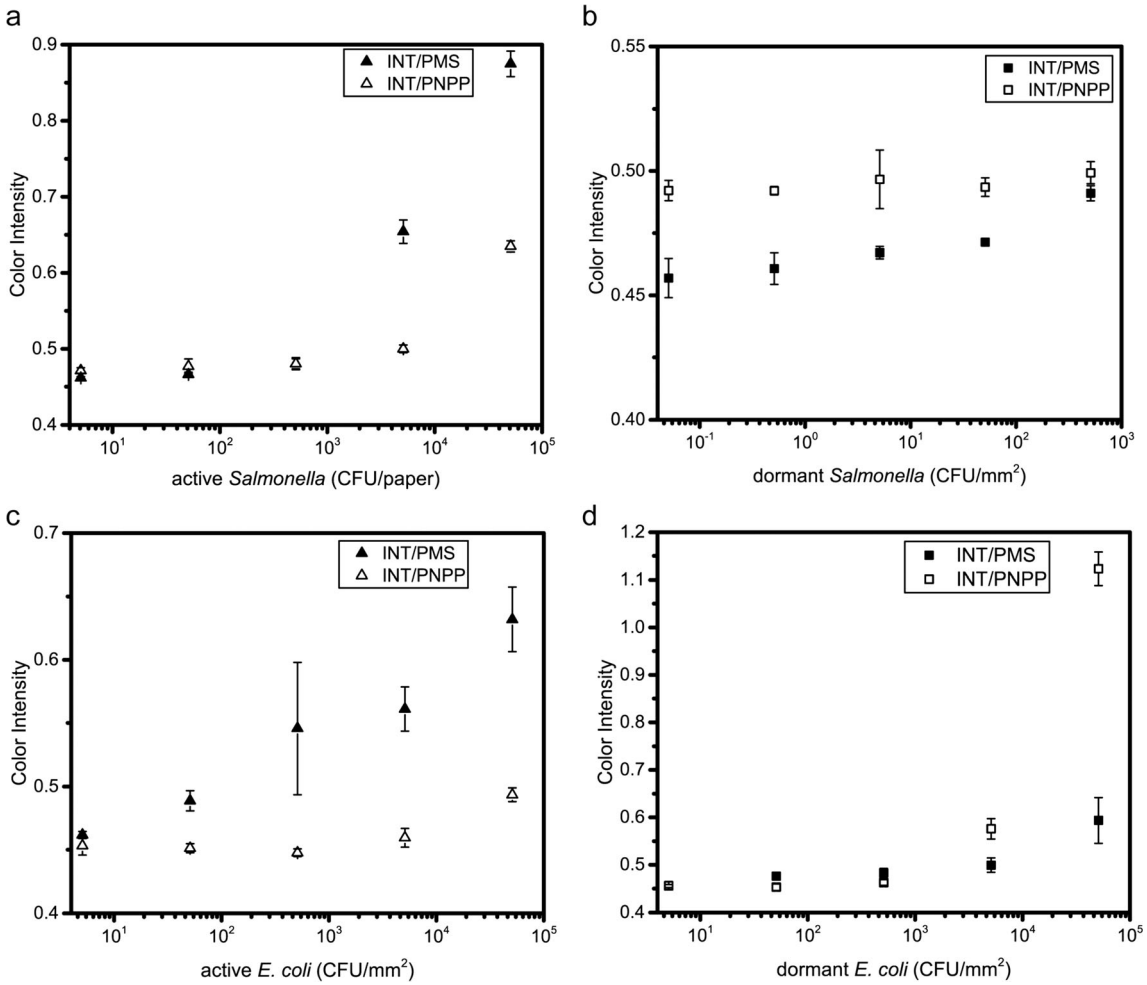

Fig. 3 Identification of bacterial metabolisms by colorimetric assays on paper. **a**, **c** Metabolically active cultures *Salmonella* Typhimurium (**a**) and *E. coli* (**c**) reacted with INT/PMS (closed triangles) and INT/PNPP (open triangles) solutions on paper. **b**, **d** Dormant *Salmonella* Typhimurium (**b**) and *E. coli* (**d**) reacted with the INT/PMS (closed squares) and INT/PNPP (open squares) on paper. Reported concentrations are 38.5 times higher if expressed per paper. INT/PMS values: green component of the color, INT/PNPP values: blue component of the color. Error bars correspond to two standard errors ($n = 5$)

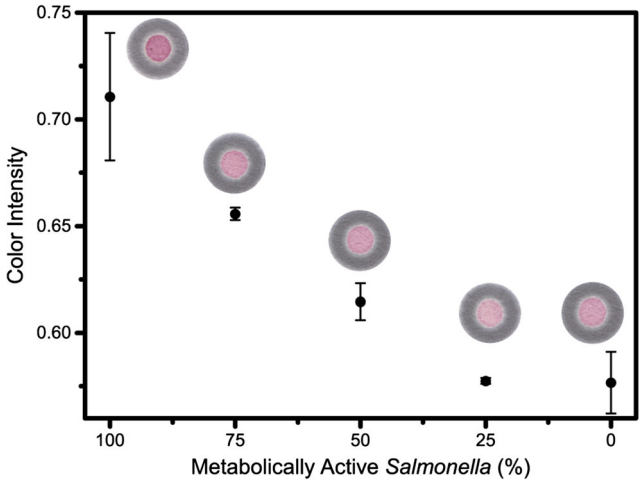

Fig. 4 INT/PMS assay on paper detected mixtures of metabolically active and dormant *Salmonella* at 10^8 CFU/ml. The green component of the purple color was used for analysis. Error bars correspond to two standard errors ($n = 5$). Inserts: representative images of the color on paper. Diameter of the circles is 5 mm

to defined areas of the paper. We used bacteriophages adsorbed on the paper to achieve specificity. T4 bacteriophages were used for *E. coli*, while P22 bacteriophages were used for *Salmonella*. T4 bacteriophages were stable to washing losing about 20% of active bacteriophages.

T4 bacteriophage adsorbed on filter paper bound dormant *E. coli* and changed color in the presence of INT/PNPP. Papers immersed for 48 h in the bacteriophage solution showed the darkest purple with INT/PNPP after *E. coli* capture. A shorter time of adsorption reduced the color intensity, suggesting a lower number of captured bacteria. Washing the papers with the adsorbed T4 bacteriophages up to eight times before capturing the bacteria did not reduce the color intensity of bound *E. coli.* The color only changed with the number of *E. coli* captured on the paper, with stronger colors for higher numbers of *E. coli* (Fig. 5). P22 bacteriophages adsorbed on paper

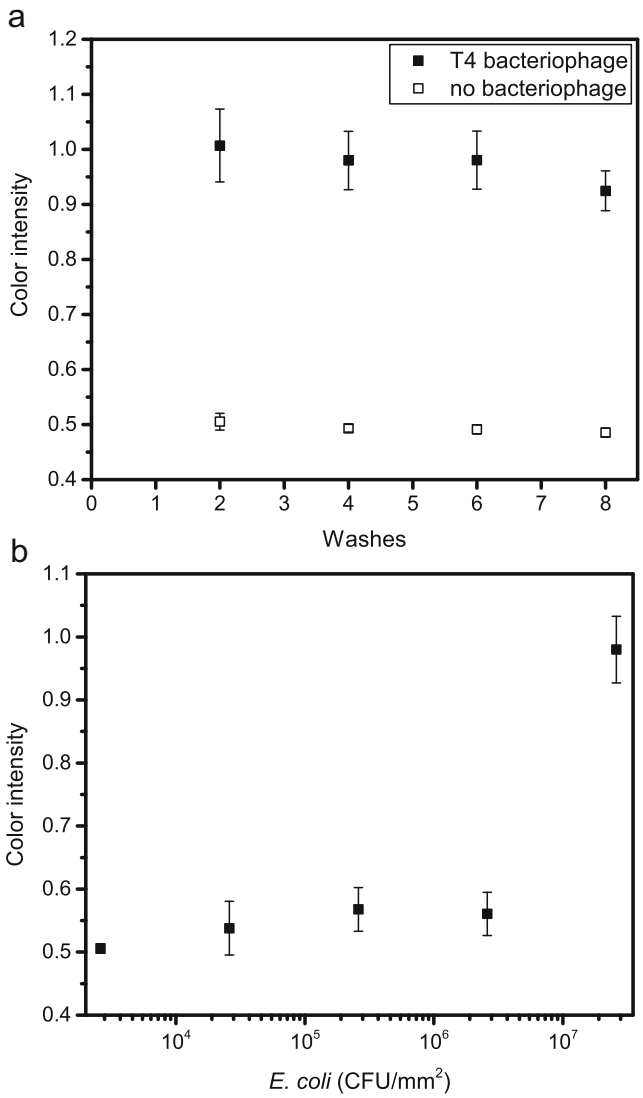

Fig. 5 Colorimetric detection of dormant *E. coli* bound to T4 bacteriophage on filter paper. **a** Change in the blue component of the color depending on the number of washes with PBS after capture of dormant *E. coli* and before analysis with INT/PNPP (3 mM/45 mM). Error bars correspond to two standard errors ($n = 8$). **b** Change in the blue component of the color depending on the concentration of dormant *E. coli*. *E. coli* were bound to bacteriophages and unbound bacteria were removed by four washes with PBS buffer before INT/PNPP (3 mM/45 mM) solution was added. Error bars correspond to two standard errors ($n = 8$)

captured metabolically active *Salmonella* that changed color with INT/PMS as low as 3 CFU/mm^2.

Discussion

Both beneficial and pathogenic bacteria can switch their metabolic states depending on their environment. Since these changes can alter their behavior and influence environmental and industrial processes, it is important to determine their metabolic state in a short time. In this study, we developed two paper-based assays that distinguished between metabolically active and dormant *E. coli* and *Salmonella* in water, and have the potential to be used with other bacteria, as they target classes of enzymes common in most bacteria. Compared to other paper-based assays for bacteria (Jokerst et al. 2012), our assays did not require bacterial lysis to release the oxidoreductases and alkaline phosphatases before reaction with the dyes, reducing the steps involved in the determination of the metabolic state of the bacteria. A drawback to not lysing the bacteria was the higher concentrations of *E. coli* and *Salmonella* required for the color change, as the dyes had to cross the membrane of the bacteria before reacting with the enzymes. If the determination of the metabolic state of lower concentrations of bacteria is needed, our assay could be combined with chemical or biological lysis once the bacteria are immobilized on the paper.

Metabolically active bacteria developed a purple color while reacting with INT and PMS due to the reduction of the tetrazolium salt by oxidoreductases. While in solution, INT and PMS concentrations below the toxic level—1 mM for INT reacting in *E. coli* (Hatzinger et al. 2003)—are generally sufficient to detect metabolically active bacteria (Smith and McFeters 1997; Ke et al. 2014), on paper millimolar quantities 5 to 40 times higher were needed. The need for a higher concentration likely came from an interaction of INT and PMS with the paper. Both molecules are positively charged, while the paper is negatively charged. This charge difference could attract INT and PMS to the paper fibers preventing them from interacting with the bacteria. The higher concentration of INT in our assay likely determined which oxidoreductases were present in the bacteria. Only the lowest concentration of INT tested (0.5 mM) was below the toxic level suggesting that oxidoreductases involved in normal cellular processes were mainly responsible for the color change. At higher concentrations of INT, oxidoreductases expressed in response to the toxic levels of dye could also be present, such as cytochrome oxidases (Ullrich et al. 1996). The amount of these oxidoreductases depended on the concentration of INT as observed by an increase in the color with increasing concentration of INT above the toxic level (Fig. 1).

For dormant *E. coli*, combining INT and PNPP at pH 10 led to the darkest purple color (Fig. 2). PNPP likely reacted with alkaline phosphates in the periplasm, while INT reacted with oxidoreductases in cytosol and periplasm (Ozkanca and Flint 1996; Wang et al. 2003; Yang and Metcalf 2004; Clegg et al. 2006). As with the assay for metabolically active bacteria, millimolar concentrations of INT were required for the darkest color changes. These higher concentrations suggest that some dye adsorbed on the paper and oxidoreductases present in dormant bacteria and those for detoxification reacted with the INT.

Aside from the concentration of the dyes, the concentration and species of the bacteria influenced the color. In general, higher concentrations of bacteria led to stronger colors allowing a semi-quantitative measure for the concentration of *E. coli* and *Salmonella* at each metabolic state. For example, metabolically active *E. coli* reacting with INT/PMS developed a color at lower concentrations compared to dormant *E. coli* reacting with INT/PNPP (5×10^2 CFU/mm^2 in Fig. 3c compared to 5×10^4 CFU/mm^2 in Fig. 3d). The weaker color in dormant *E. coli* was likely due to fewer expressed oxidoreductases. When comparing dormant *E. coli* to dormant *Salmonella*, the latter exhibited color change at lower concentrations (50 CFU/mm^2 in Fig. 3b), suggesting more oxidoreductases in dormant *Salmonella*. At higher concentrations, the difference in color between the assays was generally more pronounced, making it easier to distinguish between metabolically active and dormant bacteria. High concentrations of bacteria switching from a dormant to a metabolically active state, for example, are associated with cyanobacterial blooms in environmental waters that can lead to the death of wildlife (Hallegraeff 1993) or are linked to failed bioethanol productions when dormant lactic acid bacteria revert to an active metabolic state (Murphree et al. 2014) during fermentation.

Using the INT/PMS assay to detect metabolically active *Salmonella* in mixtures with dormant *Salmonella*, we found that the color intensity decreased linearly with decreasing concentration of metabolically active *Salmonella* ($\geq$ 25% in Fig. 4). This linear decrease in the color intensity, allows the determination of the ratio of active and dormant bacteria in a mixture as long as the overall concentration is constant. For mixtures with different overall concentrations of bacteria, the color intensity, which also depends on the concentration of bacteria, will have to be normalized.

The developed assays were not specific for a given bacterium, but capturing specific bacteria with bacteriophages adsorbed on the paper allowed us to determine the metabolic state solely of the bacteria of interest. Even on unmodified paper a sufficient number of bacteriophage tails were available to capture their host bacteria. The unmodified paper, however, required a longer adsorption time compared to cellulose membranes with cationic charge (Anany et al. 2011). This longer time was likely due to weaker attractive forces between the bacteriophages and the paper, which led to slower adsorption on the paper fibers. Once bacteriophages were adsorbed, they were stable to washing and able to capture their host bacteria. Dormant *E. coli* captured with T4 bacteriophages exhibited a higher limit of detection with the INT/PNPP assay compared to adsorbed *E. coli* (10^6 CFU/mm^2 in Fig. 3d versus 10^4 CFU/mm^2 in Fig. 5b), while metabolically active *Salmonella* captured with P22 bacteriophages had a lower limit of detection with the INT/PMS assay compared to adsorbed *Salmonella*. These differences in the limit of detection for papers with and without bacteriophages and for *E. coli* and *Salmonella* were likely due to differences in experimental protocols, the nature of the bacteriophages, and their interactions with the bacteria. We did not wash papers with adsorbed bacteria to control the number of bacteria per paper for calibration and control, while we removed unbound bacteria, lowering their numbers before performing the assays. T4 and P22 bacteriophages belong to two different families, while T4 bacteriophages are a member of the Myoviridae family and have a long contractile tail, P22 bacteriophages are a member of the Podoviridae family with no tail. The long tail in the T4 bacteriophages was more sensitive to environmental factors and potential denatured during the adsorption on the paper fibers. Bacteriophages also likely preferentially bound metabolically active bacteria as shown for *Salmonella* bacteriophages on gold surfaces (Fernandes et al. 2014).

In summary, we developed two paper-based colorimetric assays to distinguish between metabolic states in *E. coli* and *Salmonella*. These assays are not specific to a certain bacterial strain and adaptable to other bacterial species or cells containing oxidoreductases and alkaline phosphatases. Combining these non-specific assays with bacteriophages allowed us to determine the metabolic state of specific bacteria. Independent of the specificity, the assays were fast (< 30 min), simple, and low-cost, allowing a semi-quantitative determination of the concentration of bacteria at each metabolic state. The assays have the potential of being portable, which will allow the rapid determination of the metabolic state of bacteria, and with that can prevent negative environmental and industrial effects from rapid bacterial growth.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no competing interests.

Ethical statement This article does not contain any studies with human participants or animals performed by any of the authors.

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